Jiyuan SHEN

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EDUCATION

Nanyang Technological University, Singapore 08/2022 - 07/2024

M.Eng. in Computer Science and Engineering, supervised by Prof. LAM, Kwok Yan GPA: 4.17/5.0

Shanghai University, Shanghai, China 09/2018 - 06/2022

B.Eng. in Intelligent Science and Technology GPA: 3.54/4.0

WORK EXPERIENCE

SAP AI Lab, Data Scientist | Singapore 07/2024 - Present

Role Summary: Architected and scaled SAP's enterprise AI evaluation and optimization platform while leading applied research.

1. **Benchmark Agent Harnesses Memory Mechanism on Long-horizon Task (unified-harness-evaluation)**
   Develop a scalable benchmark for evaluating 18 popular harnesses on curated long-horizon enterprise tasks. I am responsible for designing the unified harness runner interface, leading agent memory-behavior analysis and synthetic data generation.
2. **Skill Evaluation and Optimization**
   Reframe skill evaluation as a system-level attribution, introducing capability correctness as atomic operation to assist optimize new skills. I mainly contribute to skill eval metric integration and skill optimization.
3. **Recursive Agent Self Improvement on Hard Edge Case**
   Develop a closed loop optimization system that generates challenging cases from the environment, defines targeted evaluation metrics, and drives agent self-improvement, comparing against SFT and RL method. I mainly contribute to meta-harness improvement part.
4. **In-house Agent Evaluation Engine (agent-quality-inspect)**
   Build an end-to-end evaluation platform supporting ~200 agents across online and offline settings. Unify tracing and logging with OpenTelemetry and MLflow, integrate evaluation into CI/CD pipeline, launch a centralized leaderboard, and develop automated test-case generation tools for business units. Support daily development for hundreds of developers and product releasement.
5. **SAP LLM Chat Arena (SAP AI Core evaluations)**
   Evaluate newly released LLMs on internal scenarios, e.g., structured output (text-to-JSON), safety, document information extraction, routing, Q&A. I am responsible for collecting dataset from LoBs, maintaining internal dashboard and running pipeline.

SELECTED PUBLICATION

H Abichandani, P Chong, **Jiyuan Shen**, "EDGEGEN: Improving Tool-Calling Agents Beyond Happy Paths with Synthetic Edge Case Generation", *arxiv*

P Chong, **Jiyuan Shen**, "Talk, Evaluate, Diagnose: User-aware Agent Evaluation with Automated Error Analysis", *ICLR, 2026*

Jiyuan Shen, "OCR or Not? Rethinking Document IE in the MLLMs Era with Real-World Large-Scale Datasets", *EACL, 2026 (Oral)*

Jiyuan Shen, "Towards Efficient and Certified Recovery from Poisoning Attacks in Federated Learning ", *TIFS*

Jiyuan Shen, "A Survey on Federated Unlearning: Challenges, Methods, and Future Directions", *ACM Computing Survey*

Jiyuan Shen, "Effective Intrusion Detection in Heterogeneous IoT Networks via Ensemble Knowledge Distillation FL", *ICC, 2024 (Oral)*

SELECTED PROJECT

LLM for Science: An Intelligent QA System 10/2023 – 01/2024

- We adapted the open-source LLAMA 7b model to suit the Chinese semantic environment. We constructed 60 billion tokens SFT instruction fine-tuning dataset, incorporating alpaca-zh and scientific QA data. We applied LoRA for supervised fine-tuning.
- We employed Retrieval Augmented Generation to obtain rich context-based answers. We built an encyclopedia database and used a multi-route recall strategy to find the knowledge content most relevant to the question. Our model offers a well-adapted experience for Chinese language environments, ensuring fluent scientific Q&A interactions.

Fast Recovery from Attacks in Federated Learning 08/2023 – 01/2024

- We conducted a comprehensive survey in the promising field of Federated Unlearning. We introduce an efficient and certified method designed for fast recovery of an accurate global model from poisoning attacks in federated learning. It can reduce huge memory consumption and computational cost with only a marginal trade-off in performance.

SKILLS & QUALIFICATIONS

- **Programming Language**: Python (advanced for DS, ML), Pytorch (advanced for DL), MySQL, C++
- **Statistics**: Bayesian Learning, EM algorithm, PCA, regression, decision tree, clustering, ensemble learning, text analytics, etc.
- **ML/DL**: Transformer (bert, ViT), Federated Learning, VAE, GAN, YoloV5, Faster R-CNN, U-Net, Knowledge Distillation, etc.